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A PICTORIAL GUIDE TO THE FRESHWATER FISHES OF DAGANA



DIVISIONAL FOREST OFFICE, DAGANA

2025



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Foreword



Bhutan, nestled within the heart of the Eastern Himalayan biodiversity hotspot, stands as a paragon on a global scale of conservation. Our nation's steadfast commitment to environmental stewardship is reflected in the designation of over half our territory as protected areas, interconnected through an extensive network of national parks and biological corridors. This landscape-level approach is a testament to our unique philosophy of harmonizing conservation with national development.

While we have made significant advances in safeguarding terrestrial ecosystems, our aquatic ecosystems – so vital to the country's ecological and economic well-being – have historically received less attention. As a major watershed within the Indo-Gangetic region, Bhutan is blessed with an intricate network of rivers, streams, lakes, and reservoirs that support a remarkable diversity of freshwater life. However, these ecosystems face increasing pressures from land use changes, hydropower development, and the absence of comprehensive baseline data, raising concerns about their long-term health and resilience. It is in this context that I am pleased to present this publication – the first of its kind for the Divisional Forest Office, Dagana since its inception in November 2016. This pioneering effort by the team, marks a significant milestone in documenting the district's freshwater fish diversity and provides an essential foundation for informed conservation planning. The findings from this study will support the development of targeted management interventions to address emerging threats such as climate change and water pollution.

I commend the team from the DFO, Dagana for their dedication and foresight in initiating this important work. I encourage continued collaboration, research, and innovation to further strengthen the conservation of our freshwater ecosystems. May this publication inspire similar assessments across the country and contribute meaningfully to country's broader biodiversity conservation goals.



(Karma Tenzin)

Director

Department of Forests and Park Services

Preface



The rich and diverse freshwater ecosystems in Bhutan harbor a unique assemblage of fish species, many of which remain relatively undocumented outside scientific circles. Recognizing the ecological and cultural value of these aquatic resources, *A Pictorial Guide to the Freshwater Fishes of Dagana* was developed as an accessible and informative reference for conservationists. This guidebook serves as a visual and descriptive catalogue of the fish species found within the rivers, streams and other waterbodies under Dagana Dzongkhag. It aims to support awareness, identification, and appreciation of freshwater biodiversity, and ultimately contribute to the conservation of aquatic habitats. Designed with both professionals and non-specialists in mind, the guide provides key information in a simplified format, supported by photographs and local context. The information presented in this guide was collected through extensive fieldwork carried out across various water bodies in Dagana between March 2025 and July 2025. The species identification was conducted using standard taxonomic references and validated through expert consultations.

This publication – the very first of its kind for the Division – would not have been possible without the unwavering support of the Department of Forests and Park Services (DoFPS), the dedicated team from the Division, and all our conservation partners. We extend our sincere gratitude to those who supported in coming up with this publication. Special thanks also go to the subject experts and reviewers who contributed their insights and helped improve the accuracy and clarity of the content.

We hope this guide inspires its readers to value the freshwater biodiversity of Dagana and motivates collective efforts to protect these vital ecosystems for generations to come.

A handwritten signature in blue ink, appearing to read 'Ugyen Tshering', with a long horizontal line extending from the end of the signature.

(Ugyen Tshering)

Chief Forestry Officer

Acknowledgement

We would like to extend our heartfelt gratitude to all the individuals involved in the development of this publication of *A Pictorial Guide to the Freshwater Fishes of Dagana*. Without their active engagement and dedication, the successful completion of this endeavor would not have been possible.

We are also pleased to acknowledge Mr. Ugyen Dorji, a highly proactive Assistant Professor at the College of Natural Resources (CNR), and his dedicated team for their vital collaboration. Their involvement in the ongoing fish survey along the Sunkosh River for the Gelephu Mindfulness City (GMC) project allowed us to collectively explore and document fish diversity in the tributaries of the Sunkosh River. Mr. Ugyen has been a key contributor, offering continuous guidance and expert input throughout the process. His knowledge and commitment have greatly enriched the quality and realization of this publication.

To all who contributed — your support has been invaluable, and we thank you sincerely.

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How to Use This Guide

A Pictorial Guide to the Freshwater Fishes of Dagana has been designed to be intuitive, informative, and user-friendly, whether you are a field practitioner, student, researcher, or simply someone interested in the freshwater fishes of Dagana.

Each fish entry follows a standardized layout that allows for quick identification and comparison across species. The information is presented in both visual and descriptive formats to make it accessible to a broad audience.

Each species page includes the following sections:

- **Scientific Name:** The Latin name, including authority and year.
- **Synonym:** Alternate latin name, if any.
- **Common Name:** The English name and local/Dzongkha name (if known). Common name are coded with Eng for english, Dz for Dzongkha, Sh for Sharchopkha, Kh for Khengkha and Lh for Lhotshamkha.
- **Photograph:** An image of the fish captured from the field.
- **Description:** Key traits such as color, body shape, and fin arrangement, size, and merismetic and morphometric data
- **Habitat:** The natural environment where a freshwater fish lives, feeds, breeds, and shelters.
- **Distribution:** Specific rivers, streams, or areas in Dagana where the fish has been recorded.
- **Conservation Status:** IUCN Red List category and local threats (DD for Data Deficient, LC for Least Concern, NE for Not Evaluated, NT for Near Threatened, EN for Endangered)

To support readers who may not be familiar with technical terms, a glossary is provided at the end of the guide.

Introduction

Bhutan, located in the Eastern Himalayas, covers an area of approximately 38,394 square kilometers and is characterized by a remarkable altitudinal gradient from subtropical lowlands to alpine zones (Facts and Details, 2022). The country lies within two major biogeographic realms – the Indo-Malayan and Palearctic – which contribute to its rich biodiversity and status as a global biodiversity hotspot (National Biodiversity Centre, 2017). Forests cover 69.71% of Bhutan's land area. Over 52% of the country's land is under formal protection, supporting diverse ecosystems and more than 11,000 species of plants and animals. This extensive forest cover and the country's abundant rivers, streams, and lakes create critical habitats for a wide range of terrestrial and aquatic species, making Bhutan an important center for biodiversity conservation (FMID, 2023; National Biodiversity Centre, 2017).

Research on freshwater biodiversity in Bhutan has evolved significantly, beginning with early fish specimen collections in the 19th century to more recent comprehensive surveys documenting over 100 fish species, including native, endemic, and introduced taxa. Studies by Gurung & Thoni (2015) and Dorji & Gurung (2017) have expanded the known fish diversity, with ongoing efforts to integrate morphometric and molecular methods for accurate species identification. These taxonomic studies are essential to address increasing pressures on aquatic ecosystems from hydropower development, pollution, and climate change, and to guide conservation and sustainable management strategies (NRCRLF, 2017).

Within Bhutan, Dagana exemplifies the country's rich biodiversity, with 90.50% forest cover comprising broadleaved and coniferous species (FMID, 2023). With wide array of the altitudinal set ups, Dagana is drained by perennial rivers such as the Sunkosh, Raidak, Kalikhola, and Dagachhu, which sustain diverse freshwater fish communities including ecologically and culturally significant species such as the Golden Mahseer, and Snowtrout. These freshwater resources are vital for local livelihoods, food security, and cultural practices, highlighting the ecological and socio-economic importance of the district within Bhutan's broader landscape.

Studying freshwater fishes in Bhutan, particularly in biodiversity-rich areas like Dagana, is critical due to their multifaceted roles. Freshwater fishes maintain ecological balance by regulating aquatic food webs and contribute to nutrient cycling and water quality (Dallaire et al., 2023). They serve as bioindicators of ecosystem health, reflecting the impacts of environmental changes (WWF, 2023). Furthermore, freshwater fishes provide essential protein and economic benefits to local communities, while also holding cultural and spiritual significance (Altowairqi & Shafi, 2024). Given the threats from habitat loss, pollution, overfishing, and climate change, conserving Bhutan's freshwater fish diversity is vital for sustaining ecosystem functions and human well-being (Dallaire et al., 2023; WWF, 2023).

Objectives

- To inventory and document the diversity of fish species found in Dagana's freshwater ecosystems
- For easy identification of species for professionals, students, and citizen scientists
- To contribute to conservation planning and sustainable management of freshwater fish resources

General Morphology

The general morphology of fishes includes key external features that aid in their movement, survival, and adaptation. Most fishes have a streamlined, fusiform body divided into the head, trunk, and tail. The head includes the mouth, eyes, nostrils, and operculum (gill cover). Fishes possess paired fins – pectoral and pelvic – for steering and balance, and unpaired fins – dorsal, anal, and caudal – for stabilization and propulsion. Their body is usually covered in scales (cycloid or ctenoid in most bony fishes), and a lateral line runs along each side to detect water vibrations. Mouth position (terminal, superior, or inferior) varies based on feeding habits. Coloration serves for camouflage or communication, while body shape can differ depending on habitat and lifestyle.

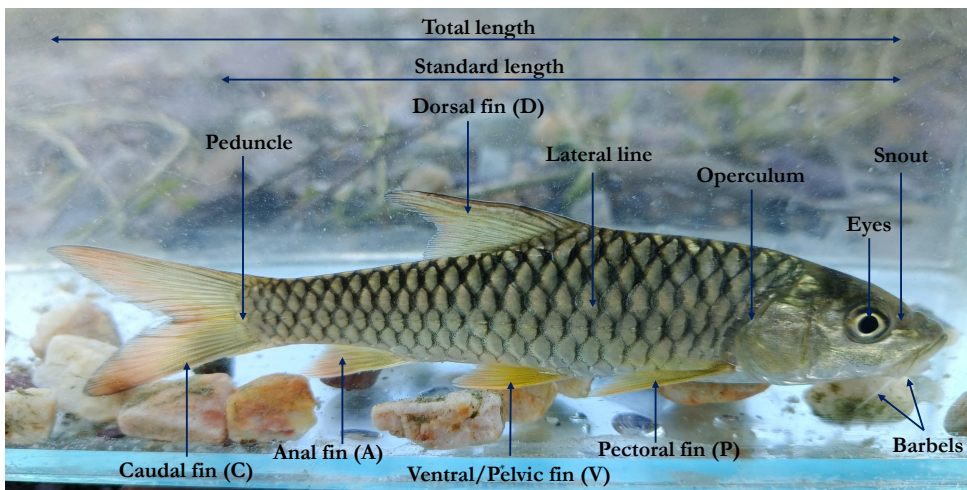


Figure 1. General morphology and body measurement of a fish

Freshwater Fishes under Dagana

The survey conducted from the month of March 2025 – July 2025 and other opportunistic encounters and social media engagement yielded the following species of fishes under Dagana:

Scientific Name: *Garra birostris* (Nebeshwar & Vishwanath, 2015)



Common Name: Buduna (Lh), Ngaripa (Kh)

Conservation Status: DD

Description: Has a prominent bilobed proboscis, each lobe bearing large, spiny acanthoid tubercles, and a transverse lobe with additional tubercles and a deep groove. It features a black spot near the gill opening and six black stripes on the caudal peduncle. The fin ray counts includes dorsal fin with 2 simple and $8\frac{1}{2}$ branched rays (D.2/ $8\frac{1}{2}$), pectoral fin with 1 simple and 12 – 15 branched rays (P.1/12–15), pelvic fin with 1 simple and 8 branched rays (P.1/8), and anal fin with 2 simple and $5\frac{1}{2}$ branched rays (A.2/ $5\frac{1}{2}$). Scale counts include 10 – 11 predorsal scales, a complete lateral line with 33–34 scales, and 16 circumpeduncular scale rows.

Habitat: Prefers clear, slightly basic water with a pH range of 7.5-8.08 and well-oxygenated river.

Distribution: Kalikhola, Samakhola, and Nirkhola

Scientific Name: *Garra annandalei* (Hora, 1921)



Synonym: *Garra chaudhuryi* Hora, 1921; *Garra satyendranathis* Ganguly & Datta, 1974;

Common Names: Stone roller (Eng), Chuche buduna (Lh), Ngaripa (Kh)

Conservation Status: LC

Description: Body elongated and cylindrical with slightly depressed head. Mouth small, semi-circular and well developed. Proboscis absent and barbels two pairs (maxillary and rostral). Interorbital space convex and upper lips ciliate. Pectoral fin shorter than head and caudal fin deeply emarginated. Body grayish, dark grey on dorsum and paler on belly. A black spot present on upper angle of gill-opening. Lower lobe of caudal black and other fins blackish. Dorsal-fin rays 9-10 (D.9–10), pectoral 15 (P.15), ventral 8 (V.8), anal 10-12 (A.10–12), and caudal 19 (C.19). Lateral scales 38-42. The body is silvery-yellowish with a brown upper back; the mouth is inferior with a blunt snout; no barbels. Maximum length is 15 cm.

Habitat: Clear, fast-flowing mountain streams with slightly basic water conditions (pH 7.5 to 8.08)

Distribution: Kalikhola, Nirkhola, Samakhola and Dareykhola

Scientific Name: *Garra arupi* (Nebeshwar, Vishwanath, & Das, 2009)



Common Name: Stone sucker (Eng), Buduna (Lh), Ngaripa (Kh)

Conservation Status: DD

Description: Characterized by a body with a shallow transverse groove on the snout tip and a band of 6–9 prominent horny tubercles on the snout. The body coloration is generally brown, and it features a distinctive black submarginal band on the dorsal fin and a longitudinal marking on the caudal fin. The fin formula includes dorsal fin rays of 2 spines and 6 soft rays (2/6), pectoral fin with 2 spines and 10–12 soft rays (2/10–12), ventral fin with 1 spine and 7–8 soft rays (1/7–8), and anal fin with 1 spine and 4–6 soft rays (1/4–6).

Habitat: Inhabits the freshwater streams and rivers, preferably the habitats with pebbly or sandy bottoms.

Distribution: Samarkhola, Nirkhola, and Darey khola

Scientific Name: *Garra quadratrostris* (Nebeshwar & Vishwanath, 2013)



Common name: Kangrey Buduna (Lh), Ngaripa (Kh)

Conservation status: DD

Description: Characterised by a dark grey body with a pale belly and a smooth body surface with a pointed snout. The fin ray counts include 11 dorsal fin rays (D.11), 15 pectoral fin rays (P.15), 8 ventral fin rays (V.8), and 17 caudal fin rays (C.17). The lateral line has 33 to 34 scales (L.l.33–34), and there are 7 lateral transverse scales (L.tr.7).

Habitat: Inhabits hill streams at altitudes of 500 to 600 meters above sea level. It is known from freshwater habitats, particularly in fast-flowing streams with rocky substrates.

Distribution: Samakhola, Dagachhu, Budichhu, and Sunkosh.

Scientific Name: *Garra gotyla* (Gray, 1830)



Common Name: Sucker head (Eng), Buduna (Lh), Ngaripa (Kh), and Bobenang (Sh)

Conservation Status: LC

Description: Has an elongated body with a prominent proboscis featuring a shallow median depression, giving it a bilobed appearance. The upper lip bears a thin band of papillae arranged in two ridges. Two pairs of barbels are present. The fin formula includes a dorsal fin with 31 rays (D.31), pectoral fin with 12 rays (P.12), ventral fin with 8 rays (V.8), anal fin with 15 rays (A.15), and a caudal fin with 17 rays (C.17). The lateral line scale count is around 34. The body color is typically dark grey with silvery scales.

Habitat: It inhabits in the fast-flowing rivers and streams.

Distribution: Samakhola, Dareykhola and Nirkhola

Scientific Name: *Garra parastenorhynchus* (Thoni, Gurung & Mayden, 2016)



Common Name: Buduna (Lh), Ngaripa (Kh)

Conservation Status: NE

Description: Characterized by a prominent, overhanging club-shaped proboscis, which is a distinctive morphological feature. Its body coloration is dusky green on the dorsal side, golden along the flanks, and greenish-white on the ventral side. It possesses two pairs of barbels. The fin ray counts include 8 dorsal fin rays (D.8), 14 pectoral fin rays (P.14), 7 ventral fin rays (V.7), 5 anal fin rays (A.5), and 19 caudal fin rays (C.19). The lateral line has 32 scales (L.l.32).

Habitat: Inhabits runs with cobblestone substrates in low-current sections of rivers.

Distribution: Samakhola and Nirkhola

Scientific Name: *Crossocheilus latius* (Hamilton, 1822)



Synonym: *Cirrhhina latia* (Hamilton, 1822); *Cyprinus latius* Hamilton, 1822; *Cyprinus gobama* Hamilton, 1822; *Cyprinus sada* Hamilton, 1822; *Tariqilabeo latius* (Hamilton, 1822)

Common Name: Gangetic latia (Eng), Minor carp (Eng), Sunkoshi buduna (Lh), Ngaripa (Kh)

Conservation Status: LC

Description: Has an elongated body with a relatively small head and an inferior mouth. The upper lip is fringed and biofilm from submerged surfaces. It typically has one pair of rostral barbels, although some individuals may possess two pairs. The fin ray counts include 10 to 11 dorsal fin rays, 15 pectoral fin rays, 9 ventral (pelvic) fin rays, 7 anal fin rays, and 19 caudal fin rays. The maximum length is approximately 12.5 cm.

Habitat: Inhabits clear, fast-flowing hill streams with rocky or gravelly substrates and is distributed in hill streams and rivers.

Distribution: Cherichhu/Salamji, Samakhola, and Sunkosh.

Scientific Name: *Psilorhynchus homaloptera* (Hora & Mukerji, 1935)



Synonym: *Psilorhynchus homaloptera rowleyi* Hora & Misra, 1941

Common Names: Balitora Minnow (Eng), Tite Buduna (Lh), Ngaripa (Kh)

Conservation Status: LC

Description: Characterized by a depressed head and a ventrally placed mouth. It lacks barbels, which is typical for this genus. Its body is marked with blotches and has a lateral scale count ranging from 30 to 35. The fin ray counts are dorsal fin 9 – 10, pectoral fin 17, ventral fin 9, anal fin 2 simple and 5 branched rays, and caudal fin 18 rays. The maximum standard length reaches about 7 cm.

Habitat: It is native to the high-gradient streams and inhabits in fast-flowing, rocky-bottomed streams.

Distribution: Labrang khola, Sabrang khola, Homa khola, and Sama khola.

Scientific Name: *Balitora brucei* (Gray, 1830)



Synonym: *Balitora maculata* Gray, 1830; *Homaloptera brucei* (Gray, 1830); *Homaloptera maculata* (Gray, 1830); *Platy cara maculata* (Gray, 1830); *Platy cara anisura* McClelland & Griffith, 1842; *Balitora brucei burmanicus* Hora, 1932

Common Name: Gray's stone loach (Eng), Tite Buduna (Lh)

Conservation Status: NT

Description: Has a streamlined bod and distinctive fin ray counts: 3 spines and 8 soft rays in the dorsal fin (D.3/8), 21 pectoral rays (P.21), 11 ventral rays (V.11), 6–8 anal rays (A.6–8), and 17 caudal rays (C.17). Its presence is an indicator of clean, oxygen-rich water, as the species is sensitive to pollution and habitat disturbance..

Habitat: Well-adapted to the fast-flowing and rocky streams.

Distribution: Homa khola and Sabrangkhola

Scientific Name: *Opsarius barna* (Hamilton, 1822)



Synonym: *Barilius barna* (Hamilton, 1822); *Cyprinus barna* Hamilton, 1822; *Opsarius fasciatus* McClelland, 1839; *Opsarius latipinnatus* McClelland, 1839; *Barilius jayarami* Barman, 1985; *Barilius barana* (Hamilton, 1822); *Lenciscus acanthopterus* (McClelland, 1839)

Common Name: Barna baril (Eng), Tite kane fageta (Lh)

Conservation Status: LC

Description: It has silvery body marked with 9 to 11 vertical dark blue or black bands crossing the lateral line, which consists of 39 – 42 scales. It lacks barbels. The fin ray counts are dorsal fin 2 simple + 7 branched rays, pectoral fin 15 rays, ventral fin 9 rays, anal fin 3 simple + 10 – 11 branched rays, and caudal fin 19 rays. The species grows up to about 10 – 13 cm in length. It is a fast-swimming, surface-feeding fish that preys mainly on aquatic and terrestrial insects, with some small fishes and benthic invertebrates likely included in its diet. It prefers clear, cool, and flowing waters and is often found in hill streams and larger rivers.

Habitat: Inhabits well-oxygenated, medium-to-high gradient rivers and streams with substrates of gravel, cobbles, boulders, and exposed bedrock, often in fast-flowing or torrential waters.

Distibution: Jigmechhu, Homa Khola, and Kalikhola

Scientific Name: *Barilius bendelisis* (Hamilton, 1807)



Synonym: *Cyprinus bendilisis* Hamilton, 1807; *Cyprinus chedra* Hamilton, 1822; *Cyprinus tila* Hamilton, 1822; *Leuciscus branchiatus* McClelland, 1839; *Barilius howesi* Barman, 1986; *Leuciscus chedra* (Hamilton, 1822)

Common Name: Hamilton's barila (Eng), Fageta (Lh)

Conservation Status: LC

Description: Has an elongated and compressed body with a moderately cleft mouth. Possesses two short pairs of barbels. The dorsal profile is less convex than the ventral profile, and the dorsal fin is entirely inserted ahead of the anal fin. The maximum standard length is 15.5–22.7 cm. The fin ray counts include dorsal fin rays of 9 (2 simple and 7 branched), pectoral fin rays between 13 and 15, ventral fin rays 9, anal fin rays 2–3 simple and 8 branched, and 18 caudal fin rays. The lateral scales number between 39–45. The body coloration is silvery with 10–12 dark vertical bands descending towards the lateral line. The fins often show yellowish tinges, and the dorsal fin typically has a dark edge.

Habitat: It inhabits clear, fast-flowing hill streams and rivers with rocky or pebbly bottoms.

Distribution: Samakhola, Chuni Khola, and Kalikhola

Scientific Name: *Barilius vagra* (Hamilton, 1822)



Synonym: *Cyprinus vagra* Hamilton, 1822; *Barilius alburnus* Günther, 1868; *Barilius bleekeri* Day, 1872; *Leuciscus piscatorius* McClelland, 1842; *Leuciscus vagra* (Hamilton, 1822); *Opsarius bicirratu*s McClelland, 1842; *Opsarius isocheilus* McClelland, 1839; *Opsarius piscatorius* McClelland, 1842; *Shacra vagra* (Hamilton, 1822).

Common Name: Vagra baril (Eng), Lam fageta (Lh)

Conservation Status: LC

Description: Has a slender, laterally compressed body that is silvery white in color, marked with 10–14 vertical bars or oblong spots along the flanks, each reaching about half the body depth. It possesses two pairs of barbels: the rostral pair is about half the length of the head, while the maxillary pair is very short. The fin formula includes 9 dorsal-fin rays (usually 2 simple and 7 branched), 16 pectoral rays, 9 ventral rays, 13–15 anal rays, and 19 caudal rays. The maximum length is 12.5 cm.

Habitat: It typically inhabits well-oxygenated, fast-flowing hill streams and rivers with rocky or gravelly substrates.

Distribution: Samakhola, Jigmechhu, Homa khola, and Kalikhola

Scientific Name: *Raiamas bola* (Hamilton, 1822)



Synonym: *Barilius bola* (Hamilton, 1822); *Barilius corbetti* Tilak & Husain, 1980; *Cyprinus bola* Hamilton, 1822; *Cyprinus goha* Hamilton, 1822

Common Name: Trout barb (Eng)

Conservation Status: LC

Description: It has a trout-like, slender body with silvery coloration and a golden lateral stripe. Maximum reported length is about 30–35 cm. The fin ray counts are closer to dorsal-fin rays 10-11 (D.10-11), pectoral 13 (P.13), ventral 9 (V.9), anal 13 (A.13).

Habitat: Inhabits freshwater environments, primarily in clear streams and rivers along the base of hills and mountainous regions.

Distribution: Sunkosh

Scientific Name: *Danio dangila* (Hamilton, 1822)



Synonym: *Cyprinus dangila* Hamilton, 1822, *Danio deyi* Sen, 1985, *Perilampus reticulatus* McClelland, 1839, and *Danio meghalayensis* Sen & Dey, 1985.

Common Names: Dangila danio (Eng), butte bhitti (Lh)

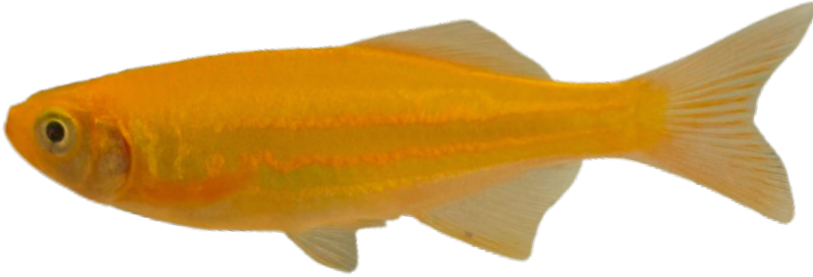
Conservation Status: Vulnerable (Vu)

Description: Has an elongated, laterally compressed body with an olive-colored back and silvery sides adorned with several narrow blue lines forming a mottled pattern. A distinctive dusky spot is present at the upper angle of the gill opening. It possesses two pairs of long barbels, giving it the common name “moustached danio.” The mouth is oblique, and the abdominal profile is more convex than the dorsal. The fin ray counts typically are dorsal fin with 11–13 rays, pectoral fin with 12 rays, ventral fin with 7 rays, anal fin with 17–18 rays, and caudal fin with 20 rays. The maximum length is about 15 cm.

Habitat: It inhabits clear, fast-flowing mountain streams. The species prefers well-oxygenated waters with rocky or stony substrates.

Distribution: Samakhola and Sabrang

Scientific Name: *Devario aequipinnatus* (McClelland, 1839)



Synonyms: *Danio aequipinnatus* (McClelland, 1839); *D. lineolatus* (Blyth, 1858); *Leuciscus aequipinnatus* (McClelland, 1839); *Leuciscus lineolatus* Blyth, 1858; *Perilampus aequipinnatus* McClelland, 1839; *Pteropsarion aequipinnatus* (McClelland, 1839)

Common Names: Giant Danio (Eng), Bhatti (Lh), Perbali (Kh)

Conservation status: LC

Description: Characterized by a slender, laterally compressed body with a silvery bluish base color adorned with yellow or orange horizontal stripes running along the flanks. These stripes may be continuous or interrupted near the head region. It has a pointed, upward-facing mouth, and barbels are either absent or very short. The fin ray counts include 12–14 dorsal fin rays, 17 pectoral fin rays, 8 ventral (pelvic) fin rays, 14–16 anal fin rays, and 19 caudal fin rays. The lateral line scales number between 32 and 34. The maximum length is about 15 cm.

Habitat: It inhabits clear, fast-flowing streams and rivers with rocky or sandy substrates.

Distribution: Chuni khola (Devitar), Jigmechhu, Labrang, Sunkosh, Homa-Sama, Dharay khola, Nirkhola, and Dagachhu

Scientific Name: *Devario assamensis* (Barman, 1984)



Synonym: *Danio assamensis* Barman, 1984

Common Name: Bhatti (Lh)

Conservation Status: Vu

Description: It exhibits the typical streamlined, laterally compressed body. It displays distinctive coloration and patterning, often with horizontal stripes or bars along its body. The fin formula includes dorsal fin rays of two spines and 11 to 14 soft rays (D.2/11–14), pectoral fin rays with one spine and 13 to 17 soft rays (P.1/13–17), ventral (pelvic) fin rays with one spine and 6 to 9 soft rays (V.1/6–9), anal fin rays with two spines and 14 to 16 soft rays (A.2/14–16), and a caudal fin with 19 rays (C.19).

Habitat: It inhabits clear, fast-flowing hill streams with rocky substrates, where it forms schools and feeds on small invertebrates and plankton.

Distribution: Sabrang khola

Scientific Name: *Tor putitora* (Hamilton, 1822)



Synonym: *Tor progeneius* (McClelland, 1839); *Cyprinus putitora* (Hamilton, 1822)

Common Name: Golden masheer (Eng), Sor (Lh), Serngya (Dz), Zeer (Kh)

Conservation Status: En

Description: Characterized by a robust body with silvery scales and distinctive reddish-yellow fins. The head length is notably longer than the body depth. It possesses two pairs of barbels. The fin formula includes 10 dorsal fin rays, 17 pectoral fin rays, 9 ventral (pelvic) fin rays, 7 anal fin rays, and 19 caudal fin rays. The lateral line typically has 28 scales.

Habitat: It inhabits fast-flowing rivers and streams. It prefers clear, oxygen-rich waters with rocky or gravelly substrates.

Distribution: Dagachhu, Sama Dovan, Homa Dovan and Sunkosh.

Scientific Name: *Neolissochilus hexagonolepis* (McClelland, 1839)



Synonyms: *Acrossocheilus hexagonolepis* (McClelland, 1839); *Barbodes hexagonolepis* (McClelland, 1839); *Barbus hexagonolepis* McClelland, 1839

Common Names: Copper Mahseer or Chocolate Mahseer (Eng), Katli (Lh), Shongnya (Kh)

Conservation Status: NT

Description: Has a robust, elongated body covered with silvery scales that exhibit a distinctive copper tinge. The head length and body depth each measure approximately $1/5$ to $1/5.5$ of the total length. It possesses two pairs of barbels and pores are sometimes present on the cheek region. The fin formula includes dorsal fin rays of 3 spines and 9 soft rays (3/9), 17 pectoral fin rays, 9 ventral (pelvic) fin rays, anal fin with 2 spines and 5 to 7 soft rays (2/5–7), and 19 caudal fin rays. The lateral line comprises 28 to 31 scales. The maximum length is about 60 cm and weighs up to 6 kg.

Habitat: It inhabits fast-flowing rivers and streams with rocky substrates.

Distribution: Sunkosh, Dagachhu, Sama khola, Homa khola, Jigmechhu

Scientific Name: *Cyprinion semiplotus* (McClelland, 1839)



Synonyms: *Cyprinus semiplotus* McClelland, 1839; *Semiplotus maclellandi* Bleeker, 1860; *Semiplotus semiplotus* (McClelland, 1839)

Common Name: Assamese kingfish (Eng), Chepti (Lho)

Conservation Status: Vu

Description: Characterized by an elongated body, reaching a maximum length of 60 cm, and features a distinctive obtuse snout with approximately 6 pores. Barbels are absent. Its fin formula includes 2–4 unbranched and 23–25 branched dorsal fin rays, 16 pectoral fin rays, 10 ventral fin rays, 2 unbranched and 7 branched anal fin rays, and 19 caudal fin rays. Lateral scales number between 27 and 33, with 6 scales above and 6 scales below the lateral line.

Habitat: It inhabits well-oxygenated, clean waters.

Distribution: Sama khola, Sunkosh, and Homa khola

Scientific Name: *Labeo dyocheilus* (McClelland, 1839)



Synonym: *Cyprinus dyocheilus* McClelland, 1839; *Gobio bicolor* McClelland, 1839

Common Name: Gardee (Lh)

Conservation Status: LC

Description: Exhibits a spindle-shaped body with a slightly more convex dorsal profile. It has two short pairs of barbels. The dorsal fin is tall, concave along the upper edge. The pectoral fin is long, reaching the ventral fin. The caudal fin deeply is forked. Fin ray counts include dorsal-fin rays 2/11 (D.2/11), pectoral 17 (P.17), ventral 9 (V.9), and anal 7 (A.7). The body coloration is generally bluish with minimal markings. Max length is 90 cm.

Habitat: Adults inhabit clear, fast-flowing large rivers and are potamodromous (migratory within freshwater)

Distribution: Sunkosh and Samakhola

Scientific Name: *Labeo boga* (Hamilton, 1822)



Synonym: *Cirrhina boga* (Hamilton, 1822); *Cyprinus boga* Hamilton, 1822; *Cyprinus falcatus* Bloch, 1795; *Gobio boga* (Hamilton, 1822)

Common Name: Boga labeo (Eng), Gardee (Lh)

Conservation Status: LC

Description: Characterized by an elongated body with a slightly convex dorsal profile and a deeply forked caudal fin. It has one pair of minute maxillary barbels and a distinctive dark spot above the pectoral fin. The species has fin ray counts of 12 dorsal, 12 pectoral, 7 ventral, 5 anal, and 39–40 caudal rays, with 39–40 lateral line scales and 13 transverse scales. Its lips are thick and roughened internally. The body coloration is generally orange with reddish tinges on the fins, which adds to its identification in the wild and local markets.

Habitat: It inhabits large rivers and their tributaries, particularly favoring areas above tidal influence.

Distribution: Sunkosh, Sama khola and Homa khola

Scientific Name: *Labeo pangusia* (Hamilton, 1822)



Synonym: *Cirrhina dyocheilus* (non McClelland, 1839); *Cyprinus pangusia* Hamilton, 1822; *Gobio bicolor* (non McClelland, 1839); *Gobio pangusia* (Hamilton, 1822) *Labeo dyocheilus* (non McClelland, 1839); *Labeo kunki* Chaudhuri, 1912; *Labeo tezpurensis* (non Chaudhuri, 1912)

Common Name: Pangusia labeo (Eng), Gardee (Lh)

Conservation Status: NT

Description: Has an elongate body with a dorsal profile more convex than the ventral. The snout is obtuse with distinct lateral lobes, and the mouth is narrow with thick, non-fringed lips. It has one pair of short maxillary barbels concealed in the labial fold. The head length is about 4.5 times in total length, and the body depth is approximately 4.5 times in total length. The fin formula consists of 2 spines and 10–11 soft rays in the dorsal fin (2/10–11), 14–16 pectoral fin rays, 9 ventral fin rays, 2 spines and 5 anal fin rays (2/5), and 19 caudal fin rays. The lateral line has about 40–42 scales. The maximum length is about 90 cm.

Habitat: It inhabits rivers, lakes, ponds, and hill streams, favoring clear, flowing waters with gravelly or rocky substrates.

Distribution: Samakhola and Sunkosh

Scientific Name: *Bangana dero* (Hamilton, 1822)



Synonym: *Chondrostoma semivelatus* Valenciennes, 1844; *Cirrhina sindensis* (non Day, 1872); *Cyprinus dero* Hamilton, 1822; *Labeo almora* (non Chaudhuri, 1912); *Labeo cephalus* Valenciennes, 1842; *Labeo dero* (Hamilton, 1822); *Labeo diplostomus* (non Heckel, 1838); *Labeo dyocheilus* (non McClelland, 1839); *Labeo henshawi* Fowler, 1924; *Labeo ricnorhynchus* (non McClelland, 1839); *Labeo rilli* Chaudhuri, 1912; *Labeo sindensis* (non Day, 1872); *Labeo soplaoensis* (non Fowler, 1934); *Similabeo dero* (Hamilton, 1822)

Common Name: Gardee (Lh)

Conservation Status: LC

Description: Characterized by a conical snout covered with numerous tubercles and a pair of short maxillary barbels. Its body coloration is dark grey with silvery scales, and it has a fin formula of dorsal fin with 12 rays (D.12), pectoral fin with 17 rays (P.17), ventral fin with 9 rays (V.9), anal fin with 8 rays (A.8), and caudal fin with 19 rays (C.19). The lateral line scale count ranges from 38 to 43.

Habitat: It inhabits swift, rocky rivers such as the Samarkhola in the Doban and Sama regions, typically at altitudes between 121 and 195 meters above sea level.

Distribution: Samakhola, Homa khola and Sunkosh

Scientific Name: *Schizothorax progastus* (McClelland, 1839)



Synonym: *Oreinus hodgsonii* Günther, 1861; *Oreinus progastus* McClelland, 1839; *Racoma progasta* (McClelland, 1839); *Schizopyge progastus* (McClelland, 1839); *Schizothoraichthys macrophthalmus* (non Terashima, 1984); *Schizothoraichthys progastus* (McClelland, 1839); *Schizothorax macrophthalmus* (non Terashima, 1984); *Schizothorax raraensis* (non Terashima, 1984)

Common Name: Dinnawah snowtrout (Eng), Chhuchey asala (Lh), Drukngya (Dz), and Maynga (Sh)

Conservation Status: LC

Description: It is characterized by a thick V-shaped sub-terminal mouth, tiny scales, and a slim, brownish body. Its fin ray counts are: dorsal fin with 11 rays, pectoral fin with 19 rays, ventral fin with 10 rays, anal fin with 2 simple and 5 branched rays, and caudal fin with 19 rays. The lateral line has 97 – 99 scales. This species can reach a maximum length of about 50 cm.

Habitat: Native to hill streams and deep run backwaters

Distribution: Samarkhola, Jigmechhu, and Nirkhola

Scientific Name: *Schizothorax richardsonii* (Gray, 1832)



Synonym: *Barbus guttatus* McClelland, 1838; *Cyprinus richardsonii* Gray, 1832; *Diptychus annandalei* Regan, 1907; *Gonorhynchus petrophilus* McClelland, 1839; *Oreinus guttatus* (McClelland, 1838); *Schizothorax nepalensis* (non Terashima, 1984); *Schizothorax richardsoni* (Gray, 1832); *Schizothorax richardsonii plagostomus* (non Heckel, 1838)

Common Name: Common snow trout (Eng), Asala (Lh)

Conservation Status: Vu

Description: Has a silvery body with black spots and fine, minute scales, along with four barbels around its mouth. Its fin ray counts are dorsal fin 11 – 12, pectoral fin 17, ventral fin 10, anal fin 7, and caudal fin 19. It is well adapted to turbulent, oxygen-rich waters, possessing a streamlined body and large, muscular paired fins that help it resist strong currents by clinging to rocks. It is primarily herbivorous, feeding on algae and detritus from the substrate. The common snow trout shows territorial behavior, defending limited food resources and shelter.

Habitat: It inhabits fast-flowing mountain rivers and streams with rocky or gravelly bottoms at altitudes ranging from 200 to 1260 meters above sea level.

Distribution: Dagachhu, Raidak, Sunkosh, Darachhu, Pelingchhu, and Budichhu

Scientific Name: *Aborichthys elongatus* (Hora, 1921)



Synonym: *Noemacheilus elongatus* (Hora, 1921)

Common Name: Gadera (Lh)

Conservation Status: LC

Description: Has a long, slender, serpentiform body marked with vertical stripes. It features a round tail with a reddish hue. The anal opening is positioned closer to the caudal fin than to the tip of the snout. The fin ray counts are dorsal fin with 8 rays (D.8), pectoral fin with 11 rays (P.11), ventral fin with 8 rays (V.8), anal fin with 7 rays (A.7), and caudal fin with 19 rays (C.19). The maximum length recorded is approximately 5.4 cm.

Habitat: It is native to fast-flowing hill streams and rivers. The species prefers well-oxygenated waters with strong currents, often inhabiting rapids and areas near waterfalls with pebble-strewn or rocky substrates.

Distribution: Kalikhola, Homa khola, and Samakhola

Scientific Name: *Batasio merianiensis* (Chaudhuri, 1913)



Synonym: *Macrones merianiensis* Chaudhuri, 1913

Common Name: Chilne macha (Lh)

Conservation Status: DD

Description: Has a slender, laterally compressed body marked with four dark vertical bands. The fin ray formula includes dorsal fin with two spines and seven soft rays (D.2/7), pectoral fin with one spine and 7–8 soft rays (P.1/7–8), ventral fin with one spine and 5 soft rays (V.1/5), anal fin with 3 to 5 spines and 8–9 soft rays (A.3-5/8–9), and caudal fin with 17 rays (C.17). The maximum recorded length is about 7.5 cm.

Habitat: It inhabits fast-flowing hill streams with rocky substrates.

Distribution: Sunkosh and Samakhola.

Scientific Name: *Acanthocobitis botia* (Hamilton, 1822)



Synonym: *Paracanthocobitis botia* (Hamilton, 1822); *Botia nebulosa* Blyth, 1860; *Cobitis botia* Hamilton, 1822; *Nemacheilus botia* (Hamilton, 1822); *Nemacheilus botia botia* (Hamilton, 1822); *Nemachilus mackenziei* Chaudhuri, 1910; *Noemacheilus botia* (Hamilton, 1822)

Common Name: Zipper Loach (Eng)

Conservation Status: LC

Description: Characterized by a slender, elongated body marked with approximately 10 dark bands along the sides. Males exhibit a suborbital groove, and a black patch is present on the caudal peduncle. Both the dorsal and caudal fins display banding patterns. It has three pairs of barbels around the mouth. The fin ray counts are dorsal fin with 12 to 17 rays (D.12–17), pectoral fin with 11 rays (P.11), ventral fin with 8 rays (V.8), anal fin with 2 spines and 5 soft rays (A.2/5), and caudal fin with 17 rays (C.17). The lateral line typically has 34 scales. Maximum length 11 cm.

Habitat: It inhabits fast-flowing, clear streams and rivers with substrates of sand, cobbles, rocks, and leaf litter.

Distribution: Sunkosh and Samakhola

Scientific Name: *Amblyceps laticeps* (McClelland, 1842)



Synonym: *Olyra laticeps* McClelland, 1842

Common Name: Catfish (Eng), Loolee (Lh)

Conservation Status: LC

Description: Has a slender, elongated body that has a depth approximately 13.3% of its standard length. Upper jaw is slightly longer than or equal to the lower jaw. The fin structure includes a dorsal fin with one spine and five soft rays (D.1/5), a ventral fin with six rays (V.6), an anal fin with nine rays (A.9), and a truncate caudal fin composed of 7+7 branched rays (C.7/7). Notably, the adipose fin extends toward the caudal fin but does not merge with it. The maximum length is 15 cm.

Habitat: It inhabits fast-flowing hill streams and clear freshwater habitats. This species is benthic and tends to hide under rocks or woody debris to shelter from strong currents.

Distribution: Kalikhola, Samakhola, and Homa Khola

Scientific Name: *Amblyceps apangi* Nath & Dey, 1989



Common Name: Lulee (Lh)

Conservation Status: LC

Description: Characterized by having equal jaws, a large and well-developed rictal fold, and smooth skin. It lacks pinnate-like rays on the outer margin of the principal caudal fin ray. Its caudal fin is truncate. It typically reaches a maximum length of about 15.9 cm. It has a prominent cup-like skin flap above the base of the pectoral spine. The species name "apangi" honors Sri Gegong Apang, a notable Indian politician. The caudal fin formula is typically described as having 7+7 branched rays, meaning there are 7 branched rays on the upper lobe and 7 on the lower lobe of the caudal fin. The caudal fin is truncate (straight-edged) rather than deeply forked.

Habitat: It inhabits fast-flowing hill streams.

Distribution: Kalikhola and Homa khola.

Scientific Name: *Pterocryptis barakensis* Vishwanath & Nebeshwar Sharma, 2006



Common Name: Barak Catfish (Eng)

Conservation Status: En

Description: Has nine branchiostegal rays and a high number of anal fin rays (65–76). Possesses a relatively shallow body depth and two pairs of barbels. A longer caudal fin, distinct sensory pores, serration patterns on the pectoral spine, and sexual dimorphism in genital papillae is the key feature of this species. The caudal fin is forked with a longer upper lobe.

Habitat: Inhabits freshwater, demersal zones of tropical hill streams. Its known habitat includes deep pools and cascades with boulder substrates at moderate altitudes.

Distribution: Sunkosh

Scientific Name: *Olyra longicaudata* (McClelland, 1842)



Synonym: *Olyra burmanica* (non Day, 1872); *Olyra elongata* Günther, 1883; *Olyra kempfi* (non Chaudhuri, 1912)

Common Name: Himalayan olyra (Eng)

Conservation Status: LC

Description: The fish has the following fin ray counts: dorsal fin with 7 rays (D.7), pectoral fin with 1 spine and 6 soft rays (P.1/6), ventral fin with 6 rays (V.6), anal fin with 23 rays (A.23), and caudal fin with 12 rays (C.12). The caudal fin is forked, with a longer upper lobe, which is characteristic of the species. Its body coloration is reddish-brown with a narrow, faint dark band along the lateral line. The pectoral fin spine is strong and serrated on both edges, aiding in anchoring the fish in fast-flowing water. Maximum

Habitat: It inhabits deep pools and cascades with boulder substrates.

Distribution: Kaijaley and Samakhola

Scientific Name: *Schistura reticulofasciata* (Singh & Banarescu, 1982)



Synonym: *Cobitis scaturigina* (McClelland, 1839); *Nemacheilus mugah* Day, 1869; *N. scaturigina* (McClelland, 1839); *N. shebbearei* Hora, 1935; *Noemacheilus scaturigina* (McClelland, 1839)

Common Name: Gadera/Gadela/Dari (Lh)

Conservation Status: LC

Description: Characterized by 9 to 12 vertical bands across its body and rounded caudal fin lobes. It typically reaches about 10 cm in length. The fin ray counts include 9–11 dorsal fin rays (D.9–11), 10 pectoral fin rays (P.10), 8 ventral fin rays (V.8), 6 anal fin rays (A.6), and 18 caudal fin rays (C.18).

Habitat: Primarily found in the pebble-bed streams. It inhabits clear, fast-flowing hill streams with rocky substrates.

Distribution: Kalikhola and Kumartar.

Scientific Name: *Schistura devdevi* (Hora, 1935)



Synonym: *Nemacheilus montanus* (Day, 1878); *Noemacheilus montanus* (Day, 1889)

Common Name: Gadera (Lh)

Conservation Status: LC

Description: Has a slender, elongated body marked by a few broad, irregular black bands that are significantly wider than the spaces between them. These bands extend from the dorsal surface down the sides but usually remain above or slightly below the lateral line. The dorsal fin is characteristically adorned with two black bands. Fin ray counts include 2 dorsal spines and 8 soft rays (D.2/8), 10 pectoral rays (P.10), 9 ventral rays (V.9), 2 anal spines and 5 soft rays (V.2/5), and 16 caudal rays (C.16).

Habitat: It is adapted to fast-flowing, clear, and oxygen-rich streams with rocky and gravel substrates.

Distribution: Found along the tributaries of Sunkosh.

Scientific Name: *Channa stewartii* (Playfair, 1867)



Synonym: *Ophiocephalus stewartii* Playfair, 1867

Common Name: Dwarf snakehead

Conservation Status: LC

Description: Has an elongated body with a bicolored pattern, predominantly brown and beige, and a flattened, snake-like head covered with large scales reminiscent of snakes. It has long dorsal fins and large mouths filled with sharp teeth. It has a fin formula typically consisting of dorsal fin rays numbering between 32 and 37 (D.32–37), pectoral fin rays around 15 to 17 (P.15–17), ventral (pelvic) fin rays usually 6 (V.6), anal fin rays ranging from 24 to 27 (A.24–27), and caudal fin rays about 18 (C.18). It grows up to about 20–25 cm in length, with a lifespan of around 15 years.

Habitat: It inhabits hill streams as well as ponds. It requires a well-structured habitat with plenty of hiding places.

Distribution: Found in the tributaries of Raidak (Kaijaley) and Sunkosh (Homa Khola), and Kalikhola.

Scientific Name: *Channa gachua* (Hamilton, 1822)



Synonym: *Ophiocephalus gachua* Hamilton, 1822; *Ophicephalus limbatus* Cuvier, 1831; *Ophicephalus marginatus* Cuvier, 1829; *Channa gachus* (Hamilton, 1822)

Common Name: Dwarf snakehead

Conservation Status: LC

Description: Has a notably large head scales and elongated dorsal and anal fins. Males typically exhibit more extended dorsal and anal fins and display more intense coloration compared to females. One of the distinctive features is the presence of white margins on the dorsal, anal, and caudal fins, giving it a striking appearance. Adults generally reach a length of 28 to 33 cm. The fin formula includes a dorsal fin rays range from 31 to 36 (D.31–36), pectoral fin rays 14 to 15 (P.14–15), ventral (pelvic) fin rays 6 (V.6), and anal fin rays 20 to 24 (A.20–24). The species has a unique scale arrangement, including 3 to 3½ scales between the lateral line and the base of the anterior dorsal rays.

Habitat: It inhabits hill streams as well as ponds. It requires a well-structured habitat with plenty of hiding places.

Distribution: Kailash and Kalikhol

Scientific Name: *Channa striata* (Bloch, 1793)



Synonym: *Ophiocephalus striatus* Bloch, 1793

Common Name: Chevron snakehead, striped snakehead, saura, hille

Conservation Status: LC

Description: It is characterized by a robust, elongated body with a dark brown coloration, marked by darker vertical bands along the flanks. The head is broad and flattened, and the mouth is large, equipped with sharp teeth suited for its carnivorous diet. The fin formula includes 37 – 46 dorsal fin rays, 17 pectoral fin rays, 6 ventral (pelvic) fin rays, 23 – 28 anal fin rays, 13 caudal fin rays, and 50 – 57 lateral line scales. Adults can attain a maximum length of up to 100 cm.

Habitat: Thrives in a variety of freshwater habitats, including stagnant lakes, streams, swamps, rice paddies, canals, ditches, ponds, and reservoirs. It is particularly common in shallow, muddy, or vegetated waters, but can also survive in open waters and endure low-oxygen conditions.

Distribution: Kajjaley, Homa khola and Kalikhola

Scientific Name: *Pseudecheneis sulcata* (McClelland, 1842)



Synonym: *Glyptosternon sulcatus* McClelland, 1842

Common Name: Sucker throat catfish (Eng), Kabre (Lh)

Conservation Status: LC

Description: Has a dark body marked with yellowish blotches, and fin ray counts include dorsal fin with 1 spine and 6 soft rays (D.1/6), pectoral fin with 1 spine and 13 soft rays (P.1/13), ventral fin with 6 rays (V.6), anal fin with 2–4 spines and 7–9 soft rays (A.2–4/7–9), and caudal fin with 17 rays (C.17). Maximum length is 20 cm.

Habitat: Prefers fast-flowing, oxygen-rich waters.

Distribution: Tributaries of Sunkosh

Scientific Name: *Glyptothorax striatus* (McClelland, 1842)



Synonym: *Glyptosternon striatus* McClelland, 1842; *Glyptosternum striatum* McClelland, 1842

Common Name: Jantaray (Lh), Catfish (Eng)

Conservation Status: NE

Description: Has a dark brown body with the adhesive apparatus on its pectoral fins. The species grows up to a maximum length of about 20 cm, with fin ray counts including 1 spine and 6 soft rays in the dorsal fin (D.1/6), 1 spine and 10 soft rays in the pectoral fin (P.1/10), 6 to 7 ventral rays (V.6–7), 11 anal fin rays (A.11), and 18 caudal fin rays (C.18).

Habitat: Prefers benthopelagic environments, meaning it lives near the bottom of the water bodies it inhabits.

Distribution: Tributaries of Sunkosh

Scientific Name: *Badis badis* (Hamilton, 1822)



Synonym: *Badis buchmanii* Bleeker, 1853; *Cyphla fasciata* (Swainson, 1839); *Labrus badis* Hamilton, 1822; *Labrus fasciata* Swainson, 1839

Common Name: Badis or dwarf chameleon fish (Eng), Sungurey, Lati macha or Limbuni macha (Lh), Phagpanya (Kh)

Conservation Status: LC

Description: It exhibits sexual dimorphism. Males are larger and more colorful, especially during breeding, displaying variegated bands of black and green across their bodies, a distinctive blotch behind the gill cover, and a row of black blotches at the base of the dorsal fin. Females are smaller and less vividly colored. It has 15–18 dorsal spines, 7–10 dorsal soft rays (D.15–18/7–10), 12 pectoral rays (P.12), 1 spine and 5 soft rays in the ventral fin (V.1/5), 3 spines and 6–8 soft rays in the anal fin (A.3/6–8), and 16 caudal fin rays (C.16).

Habitat: Inhabits slow-flowing or stagnant waters – ponds, ditches, swamps, and heavily vegetated streams, often with muddy or silty substrates and abundant aquatic plants.

Distribution: Found in Homa khola, Kaijaley, Sama khola

Scientific Name: *Pseudolaguvia shawi* (Hora, 1921)



Synonym: *Laguvia shawi* Hora, 1921; *Glyptothorax shawi* (Hora, 1921)

Common Name: Shaw's Bee Catfish or Painted Catfish (Eng)

Conservation Status: DD

Description: Has a grayish-brown body with two broad vertical bands and possesses an adhesive apparatus with a central median depression on its thorax. The fin ray counts include 5–8 dorsal fin rays (D.5–8), 8–11 pectoral rays (P.8–11), 6–8 ventral rays (V.6–8), 7–9 anal rays (A.7–9), and 17 caudal rays (C.17). It reaches a maximum length of about 3.8 cm.

Habitat: Found in mountain rapids and are known to bury themselves in sandy substrates.

Distribution: Kalikhola

Scientific Name: *Parachiloglanis hodgarti* (Hora, 1923)



Synonym: *Euchiloglanis hodgarti* Hora, 1923; *Exostoma blythii* Day, 1870; *Glyptosternum hodgarti* Hora, 1923

Common Name: Torrent catfish (Eng)

Conservation Status: LC

Description: Characterized by a depressed, elongate body with a flattened head and smooth dorsal skin, while the underside is often tuberculate. Its lips are thick, fleshy, and covered with papillae. It lacks a post-labial groove. The eyes are minute and located dorsally beneath the skin, and the gill openings are narrow, not extending below the pectoral fin base. It has homodont dentition with pointed teeth arranged in a crescent-shaped band on the upper jaw. The paired fins are modified into an adhesive apparatus. The caudal fin is emarginate to truncate. Maximum length of 6.5 cm. Fin ray counts include 1 spine and 7 soft rays in the dorsal fin (D.1/7), 1 spine and 11–12 soft rays in the pectoral fin (P1/11–12), 6 ventral fin rays (V.6), 8 anal fin rays (A.8), and 19 caudal fin rays (C.19).

Habitat: Inhabits fast-flowing mountain streams and rivers.

Distribution: Sunkosh

Scientific Name: *Pethia conchonius* (Hamilton, 1822)



Synonym: *Barbus conchonius* (Hamilton, 1822); *Puntius conchonius* (Hamilton, 1822); *Cyprinus conchonius* Hamilton, 1822; *Systomus conchonius* (Hamilton, 1822); *Barbus pyrrhopterus* (McClelland, 1839)

Common Name: Rosy barb (Eng), Nyatu (Kh)

Conservation Status: LC

Description: Dorsal-fin rays 3/8, pectoral 11-15, ventral 9, anal 7-8, and caudal 19. Has a deep, laterally compressed body with 3 dorsal spines and 8 soft dorsal rays (D.3/8), 11–15 pectoral rays (P.11–15), 9 ventral rays (V.9), 7–8 anal rays (A.7–8), and 19 caudal rays (C.19). The total length reaches up to 9 cm. It features a large black spot above the lateral line in the caudal peduncle region, and the last undivided dorsal ray is serrated. Notably, it lacks barbels. Lateral scales 24-27 and transverse 5.5/6.5. Length of head is 5 and body height 2.7 in total length

Habitat: Inhabits a variety of freshwater habitats such as ponds, rivers, and slow-flowing streams, often preferring well-vegetated areas.

Distribution: Khani khola and tributaries of Sunkosh.

Scientific Name: *Mystus dibrugarensis* (Chaudhuri, 1913)



Synonym: *Macrones montanus dibrugarensis* Chaudhuri, 1913; *Mystus montanus dibrugarensis* (Chaudhuri, 1913); *M. vittatus dibrugarensis* (Chaudhuri, 1913)

Common Name: Tengra (Eng)

Conservation Status: LC

Description: Characterized by a body marked with two prominent black blotches—one located near the tympanic (ear) region and the other at the base of the caudal fin. It lacks distinct body bands. Has an elongated body and barbels around the mouth.

Habitat: Inhabits rivers, streams, and ponds with moderate to slow currents.

Distribution: Tributaries of Sunkosh

Scientific Name: *Lepidocephalichthys guntea* (Hamilton, 1822)



Synonym: *Cobitis balgara* (Hamilton, 1822); *Canthophrys olivaceus* Swainson, 1839; *Canthophrys vittatus* Swainson, 1839; *Cobitis guntea* Hamilton, 1822; *Cobitis maya* Sykes, 1839

Common Name: Guntea loach or simply Spotted loach (En)

Conservation Status: LC

Description: Has a slender, elongated body with a greyish coloration marked by a darker band along its sides. A distinctive black patch is present at the base of the caudal fin. The species has a relatively high number of lateral scales, approximately 115. Fin ray counts include 2 dorsal spines and 6 soft rays (D.2/6), 8 pectoral rays (P.8, 7 ventral rays (V.7), 2 anal spines and 5 soft rays (A.2/5), and 16 caudal rays (C.16). The maximum recorded length is about 15 cm.

Habitat: Inhabits slow-moving or stagnant waters such as ponds, ditches, swamps, and sluggish streams, often with muddy or silty bottoms and abundant vegetation.

Distribution: Homakhola and Sunkosh

Scientific Name: *Anguilla bengalensis* (Gray, 1831)



Synonym: *Anguilla nebulosa* McClelland, 1844

Common Name: Indian mottled eel (English); Bam Masha (Lh)

Conservation Status: NE

Description: Dorsal-fin rays 250–305, pectoral 18, anal 220–250, caudal 10–12. Body elongated and snake-like with a flat, bluntly pointed tail. Skin is scaleless, mottled with numerous black spots, dorsally greenish and ventrally white. Body is covered with mucus, making it slippery. Tiny teeth present in small groups on both upper and lower jaws. Dorsal, anal, and caudal fins are fused, forming a continuous fin structure.

Habitat: Found in pools and runs with low current and boulder substrate

Distribution: Samarkhola and Nirkhola

Scientific Name: *Xenentodon cancila* (Hamilton, 1822)



Synonym: *Belone cancila* (Hamilton, 1822); *Belone graii* Sykes, 1839

Common Name: Freshwater Garfish or Asian Freshwater Needlefish (Eng), Rocket Matsa (Lh)

Conservation Status: LC

Description: Possesses an elongated, slender body adapted for swift movements in water. Its most distinctive feature is the long, needle-like jaws equipped with sharp teeth, specialized for capturing small fish and aquatic invertebrates. The fin ray counts include 16 dorsal fin rays (D.16), 11 pectoral fin rays (P.11), 6 ventral fin rays (V.6), 16 anal fin rays (A.16), and 15 caudal fin rays (C.15).

Habitat: Inhabits deep pools with substrates of cobbles and boulders, preferring slow to moderately flowing waters.

Distribution: Samarkhola, Kaijaley and Kumartar

Species Checklist

Species	Family	Conservation Status
<i>Amblyceps laticeps</i> (McClelland, 1842)	Amblycipitidae	Data Deficient
<i>Amblyceps apangi</i> Nath & Dey, 1989	Amblycipitidae	Data Deficient
<i>Anguilla bengalensis</i> (Gray, 1831)	Anguillidae	Near Threatened
<i>Badis badis</i> (Hamilton, 1822)	Badidae	Least Concern
<i>Batasio merianiensis</i> (Chaudhuri, 1913)	Bagridae	Data Deficient
<i>Ohya longicaudata</i> (McClelland, 1842)	Bagridae	Data Deficient
<i>Mystus dibrugarensis</i> (Chaudhuri, 1913)	Bagridae	Near Threatened
<i>Balitora brucei</i> (Gray, 1830)	Balitoridae	Not Evaluated
<i>Xenentodon cancila</i> (Hamilton, 1822)	Belonidae	Least Concern
<i>Channa stewartii</i> (Playfair, 1867)	Channidae	Least Concern
<i>Channa gachua</i> (Hamilton, 1822)	Channidae	Least Concern
<i>Channa striata</i> (Bloch, 1793)	Channidae	Least Concern
<i>Lepidocephalichthys guntea</i> (Hamilton, 1822)	Cobitidae	Least Concern
<i>Garra birostris</i> (Nebeshwar & Vishwanath, 2015)	Cyprinidae	Not Evaluated
<i>Garra annandalei</i> (Hora, 1921)	Cyprinidae	Least Concern
<i>Garra arupi</i> (Nebeshwar, Vishwanath, & Das, 2009)	Cyprinidae	Not Evaluated
<i>Garra quadratrostris</i> (Nebeshwar & Vishwanath, 2013)	Cyprinidae	Not Evaluated
<i>Garra gotyla</i> (Gray, 1830)	Cyprinidae	Not Evaluated
<i>Garra parastenorhynchus</i> (Thoni, Gurung & Mayden, 2016)	Cyprinidae	Not Evaluated
<i>Crossocheilus latius</i> (Hamilton, 1822)	Cyprinidae	Not Evaluated
<i>Barilius bendelisis</i> (Hamilton, 1807)	Cyprinidae	Not Evaluated
<i>Barilius vagra</i> (Hamilton, 1822)	Cyprinidae	Not Evaluated
<i>Raiamas bola</i> (Hamilton, 1822)	Cyprinidae	Least Concern
<i>Tor putitora</i> (Hamilton, 1822)	Cyprinidae	Endangered
<i>Neolissochilus hexagonolepis</i> (McClelland, 1839)	Cyprinidae	Near Threatened
<i>Cyprinion semiplotus</i> (McClelland, 1839)	Cyprinidae	Near Threatened
<i>Labeo boga</i> (Hamilton, 1822)	Cyprinidae	Data Deficient

<i>Labeo pangusia</i> (Hamilton, 1822)	Cyprinidae	Vulnerable
<i>Labeo dyocheilus</i> (McClelland, 1839)	Cyprinidae	Least Concern
<i>Bangana dero</i> (Hamilton, 1822)	Cyprinidae	Least Concern
<i>Schizothorax progastus</i> (McClelland, 1839)	Cyprinidae	Least Concern
<i>Schizothorax richardsonii</i> (Gray, 1832)	Cyprinidae	Near Threatened
<i>Pethia conchoni</i> (Hamilton, 1822)	Cyprinidae	Least Concern
<i>Opsarius barna</i> (Hamilton, 1822)	Danionidae	Not Evaluated
<i>Danio dangila</i> (Hamilton, 1822)	Danionidae	Not Evaluated
<i>Devario aequipinnatus</i> (McClelland, 1839)	Danionidae	Least Concern
<i>Devario assamensis</i> (Barman, 1984)	Danionidae	Vulnerable
<i>Aborichthys elongatus</i> (Hora, 1921)	Nemacheilidae	Data Deficient
<i>Acanthocobitis botia</i> (Hamilton, 1822)	Nemacheilidae	Least Concern
<i>Schistura reticulofasciata</i> (Singh & Bănărescu, 1982)	Nemacheilidae	Vulnerable
<i>Schistura devdevi</i> (Hora, 1935)	Nemacheilidae	Vulnerable
<i>Psilorhynchus homaloptera</i> (Hora & Mukerji, 1935)	Psilorhynchidae	Not Evaluated
<i>Pterocryptis barakensis</i> Vishwanath & Nebeshwar Sharma, 2006	Siluridae	Endangered
<i>Pseudecheneis sulcata</i> (McClelland, 1842)	Sisoridae	Data Deficient
<i>Glyptothorax striatus</i> (McClelland, 1842)	Sisoridae	Data Deficient
<i>Pseudolaguvia shawi</i> (Hora, 1921)	Sisoridae	Data Deficient
<i>Parachiloganis hodgarti</i> (Hora, 1923)	Sisoridae	Data Deficient

Glossary

Term	Definition
Barbels	Whisker-like sensory organs located near the mouth of some fish. Used to detect food in murky waters.
Caudal Fin	The tail fin of a fish, used for propulsion and steering.
Dorsal Fin	The fin located on the back (top) of a fish, used for stability during swimming.
Endemic	A species that is native to and found only within a specific geographic area.
Fin Rays	The bony or cartilaginous structures that support the fins of fish.
Forked Tail	A tail fin that is divided into two equal lobes, resembling the shape of a fork.
Gills	The respiratory organs of fish, used to extract oxygen from water.
Habitat	The natural environment where a fish lives, such as rivers, streams, or ponds.
IUCN Red List	A global inventory that assesses the conservation status of species (e.g., Least Concern, Endangered).
Lateral Line	A line of sensory organs along the side of a fish that detects movement and vibration in the water.
Morphology	The study of the form and structure of organisms.
Pectoral Fins	Paired fins located on the sides of the fish, used for balance and steering.
Pelvic Fins	Fins located on the lower part of the fish, near the belly.
Scales	The small, plate-like structures that cover the skin of most fishes.
Standard Length (SL)	The length of a fish measured from the tip of the snout to the base of the tail (excluding the caudal fin).
Total Length (TL)	The length of a fish from the tip of the snout to the end of the tail fin.

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“This pictorial guide showcases the freshwater fish diversity of Dagana Dzongkhag, serving as a tool for education, conservation, and research. It includes descriptions and vivid images to support species identification and promote awareness of aquatic biodiversity.”